

# Limits 3: To Infinity and Beyond

Numberbender | WORKSHEET



Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

Evaluate each limit at infinity. Show all work.

## LIMITS WORKSHEET #3

### TO INFINITY AND BEYOND !!!!!

|                                                        |                                                                 |                                                      |                                                              |
|--------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------|
| 1. $\lim_{x \rightarrow \infty} 5 - \frac{2}{x^2}$     | 2. $\lim_{x \rightarrow -\infty} \frac{3x+7}{5x}$               | 3. $\lim_{x \rightarrow \infty} \frac{2x-1}{x+1}$    | 4. $\lim_{x \rightarrow \infty} \frac{2x+5}{3x^2+1}$         |
| 5. $\lim_{x \rightarrow \infty} \frac{2x^2+5}{3x^2+1}$ | 6. $\lim_{x \rightarrow \infty} \frac{2x^3+5}{3x^2+1}$          | 7. $\lim_{x \rightarrow \infty} \frac{2x^2-4x}{x+1}$ | 8. $\lim_{x \rightarrow -\infty} \frac{2x^2-4x}{x+1}$        |
| 9. $\lim_{x \rightarrow \infty} \frac{x^2+2}{x^3-1}$   | 10. $\lim_{x \rightarrow \infty} \frac{x^2+2}{x^2-1}$           | 11. $\lim_{x \rightarrow \infty} \frac{x^2+2}{x-1}$  | 12. $\lim_{x \rightarrow \infty} \frac{3-2x}{3x^3-1}$        |
| 13. $\lim_{x \rightarrow \infty} \frac{3-2x}{3x-1}$    | 14. $\lim_{x \rightarrow \infty} \frac{3-2x^2}{3x-1}$           | 15. $\lim_{x \rightarrow \infty} \frac{2x-1}{3x+2}$  | 16. $\lim_{x \rightarrow \infty} \frac{3x^3+2}{2x^2-9x^3+7}$ |
| 17. $\lim_{x \rightarrow \infty} \frac{x}{x^2-1}$      | 18. $\lim_{x \rightarrow -\infty} \frac{4x^2+3}{2x^2-1}$        | 19. $\lim_{x \rightarrow \infty} 10 - \frac{2}{x^2}$ | 20. $\lim_{x \rightarrow \infty} 4 + \frac{3}{x}$            |
| 21. $\lim_{x \rightarrow -\infty} \frac{5x^2}{x+3}$    | 22. $\lim_{x \rightarrow -\infty} \frac{1}{2}x - \frac{4}{x^2}$ |                                                      |                                                              |



## Limits 3: To Infinity and Beyond

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Answer key — for instructor use only.

### Answers:

|                    |                  |                   |                    |                  |             |
|--------------------|------------------|-------------------|--------------------|------------------|-------------|
| 1) 5               | 2) $\frac{3}{5}$ | 3) 2              | 4) 0               | 5) $\frac{2}{3}$ | 6) $\infty$ |
| 7) $\infty$        | 8) $-\infty$     | 9) 0              | 10) 1              | 11) $\infty$     | 12) 0       |
| 13) $-\frac{2}{3}$ | 14) $\infty$     | 15) $\frac{2}{3}$ | 16) $-\frac{1}{3}$ | 17) 0            | 18) 2       |

